JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SHREE VARI CORPORATION

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

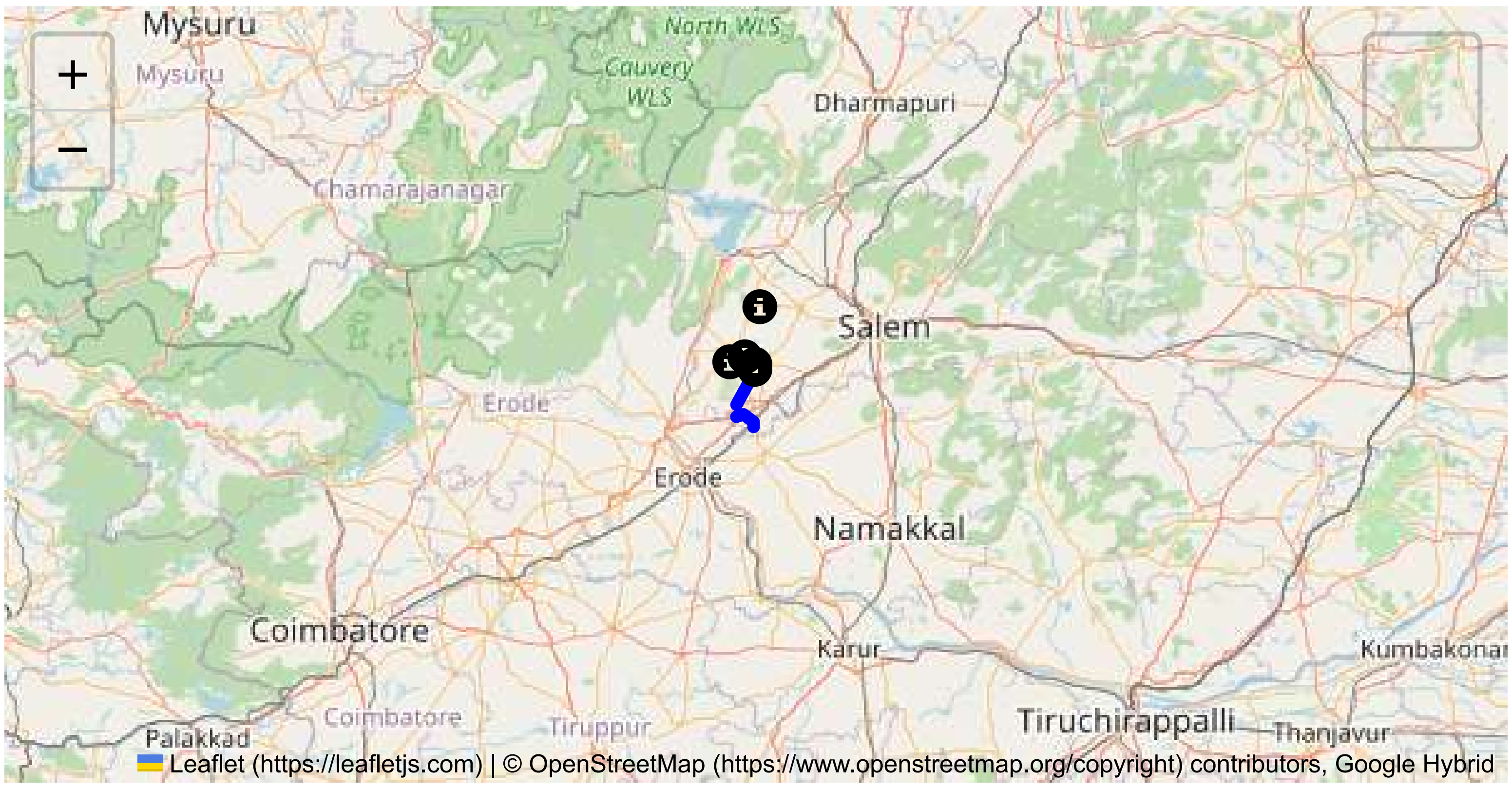
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

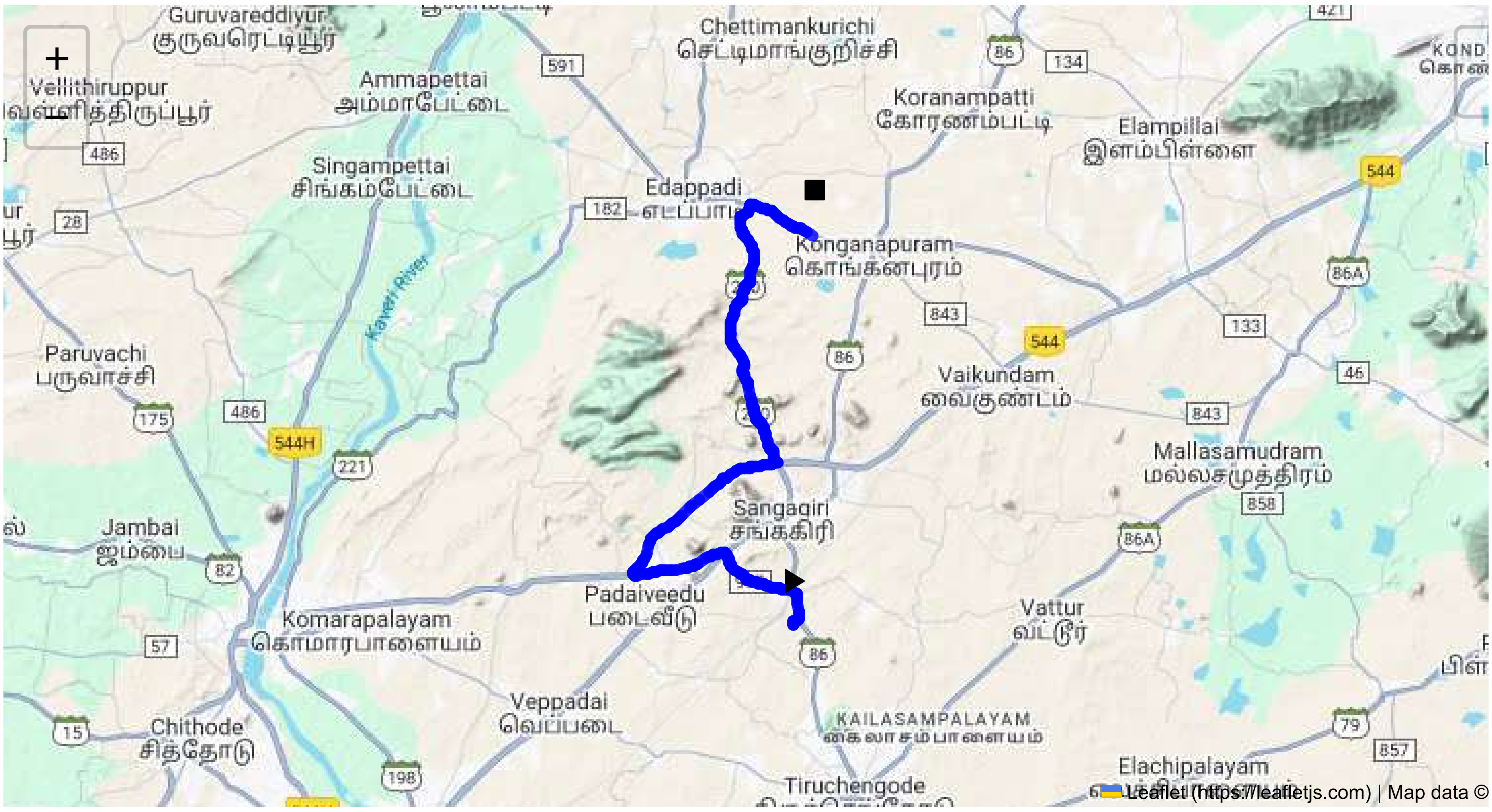
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 29.63 km
Estimated Duration: 0.6 hours
Adjusted Duration (Heavy Vehicle): 0.8 hours
Start: (11.4381, 77.8734)
End: (11.572583, 77.880157)



Welcome to the Journey Risk Management Study

Route Overview

The designated route from CVQF+23W, Sangagiri to HVFJ+634, Reddippatti spans approximately 29.63 kilometers. The path involves traveling through the Sankari RS - Erode Privu Road and Bhavani Main Road, traversing a mix of rural roads and outskirts of populated areas, particularly in Sankari and Padaiveedu.

Weather Conditions and Hazards

Tamil Nadu experiences typical tropical climate conditions. The primary weather issue could be the monsoon, with heavy rains during the southwest (June-September) and northeast (October-December) monsoon seasons resulting in waterlogging on roads. These conditions can lead to reduced visibility and slippery roads, increasing accident probability.

Traffic Patterns and Congestion

Traffic along these roads can vary, with peak congestion typically occurring in the morning (7:00-9:00 AM) and evening (5:00-7:00 PM) due to local commuters. Congestion may be expected near urban intersections like Sankari, given local commerce and transit.

Road Quality and Infrastructure

The road quality along this route is generally moderate; however, sections of the Bhavani Main Road may see wear and tear, potholes, and uneven surfaces due to heavy vehicle movement. Roadside amenities are limited outside populated areas.

Alternative Routes

For emergencies, an alternative might be taking local roads branching off major routes, though these require familiarity with local infrastructure. Ensure alternative routes are checked for load capacity and access permissions for heavy vehicles.

Local Regulations for Hazardous Material Transport

Local regulations require hazmat vehicles to operate during off-peak hours to minimize risks. Proper labeling, documentation, and adherence to load limits are crucial. Night travel might be restricted depending on the nature of materials.

Historical Incidents

Historical data indicates occasional accidents involving heavy trucks, primarily due to overloading and brake failures. There have been minor spills, indicating the need for stringent adherence to safety protocols.

Environmental Considerations

Passing through rural zones, active monitoring for wildlife crossings and maintaining ecological areas is essential. Sensitive zones should be observed, especially during night travel to avoid disturbances.

Communication Coverage

Generally stable mobile network coverage exists along major roads, though certain rural stretches may experience brief dead zones, particularly in less populated areas.

Emergency Response Times

Emergency response times can vary but generally expect faster response within 20-30 minutes around Sangagiri and more populated areas. Rural areas may have longer response times due to the distance from emergency services.

Risk Assessment Summary

The route primarily involves low to moderate risk, with increased caution needed during monsoon seasons and peak traffic hours. Ensuring vehicle compliance with local regulations and driver preparedness for potential road and weather conditions are vital. Communication systems should be reliable, and preparation for emergencies must be emphasized, especially for hazmat transport. Regular updates and familiarity with local infrastructure can greatly reduce risks significantly.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.58300, 77.85813	15 KM/Hr
1	Turn	High	11.43811, 77.87348	15 KM/Hr
2	Turn	Medium	11.43956, 77.87340	30 KM/Hr
3	Turn	High	11.43968, 77.87345	15 KM/Hr
4	Turn	High	11.44029, 77.87544	15 KM/Hr
5	Turn	Medium	11.45357, 77.85789	30 KM/Hr
6	Turn	Medium	11.45348, 77.85706	30 KM/Hr
7	Turn	Medium	11.45352, 77.85698	30 KM/Hr
8	Turn	Medium	11.46318, 77.84913	30 KM/Hr
9	Turn	High	11.45542, 77.81725	15 KM/Hr
10	Turn	High	11.45631, 77.81668	15 KM/Hr
11	Turn	Medium	11.49249, 77.85820	30 KM/Hr
12	Blind Spot	Blind Spot	11.49390, 77.86748	10 KM/Hr
13	Turn	High	11.57903, 77.85432	15 KM/Hr

Emergency Locations

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium

Route Photos of Risky Spots

Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.58300, 77.85813

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45357, 77.85789

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45348, 77.85706

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45352, 77.85698

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46318, 77.84913

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45542, 77.81725

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45631, 77.81668

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49249, 77.85820

Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.49390, 77.86748

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.57903, 77.85432